

Day 5: Network Testing Practical

Task 1 : Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Objective

To verify whether a laptop and a friend's phone connected to the same Wi-Fi network can communicate with each other using the ping command.

Devices Used

- Windows Laptop
- Android Phone

Procedure

Step 1:

Connected both the laptop and Android phone to the **same Wi-Fi network**.

Step 2:

Checked the IP address of both devices.

- **Laptop IP Address:** 192.168.0.11
- **Phone IP Address:** 192.168.0.4

Step 3:

From the laptop, opened **Command Prompt** and used the phone's IP address:

ping 192.168.0.4

The laptop received successful **Reply** messages from the phone.

Step 4:

From the phone, performed a ping test to the laptop's IP address using a Termux:

ping 192.168.0.11

The phone also received successful replies from the laptop.

Screenshots

Screenshot 1: Laptop IP address

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . . : 2406:b400:f11:e9f7:1b00:f970:c3a8:329b
Temporary IPv6 Address. . . . . : 2406:b400:f11:e9f7:5141:1b33:3d3e:3ad0
Link-local IPv6 Address . . . . . : fe80::7c57:b381:11:51d%8
IPv4 Address. . . . . : 192.168.0.11
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::bafb:b3ff:fea2:4a1b%8
                             192.168.0.1
```

```

10:06 >_                                     3.63 KB/s 🕒 🔌 📶 🔋84%
~ $ ping -c 5 192.168.0.11
PING 192.168.0.11 (192.168.0.11) 56(84) bytes of
data.
64 bytes from 192.168.0.11: icmp_seq=1 ttl=128 t
ime=12.6 ms
64 bytes from 192.168.0.11: icmp_seq=2 ttl=128 t
ime=10.7 ms
64 bytes from 192.168.0.11: icmp_seq=3 ttl=128 t
ime=11.9 ms
64 bytes from 192.168.0.11: icmp_seq=4 ttl=128 t
ime=12.0 ms
64 bytes from 192.168.0.11: icmp_seq=5 ttl=128 t
ime=10.9 ms

--- 192.168.0.11 ping statistics ---
5 packets transmitted, 5 received, 0% packet los
s, time 4006ms
rtt min/avg/max/mdev = 10.733/11.673/12.688/0.73
9 ms
~ $ █
```

```
C:\Users\milan>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:
Reply from 192.168.0.4: bytes=32 time=77ms TTL=64
Reply from 192.168.0.4: bytes=32 time=75ms TTL=64
Reply from 192.168.0.4: bytes=32 time=3ms TTL=64
Reply from 192.168.0.4: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 77ms, Average = 39ms
```

The ping test was **successful in both directions**. The laptop successfully communicated with the phone, and the phone successfully communicated with the laptop.

Task 2 : Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Objective:

To determine the number of network hops required for a connection from the local device to reach www.youtube.com.

Command Used:

tracert www.youtube.com

Procedure:

1. Opened **Command Prompt** on the Windows laptop.
2. Entered the command **tracert www.youtube.com**.
3. Pressed **Enter** and waited for the traceroute to complete.
4. Observed the list of routers/network devices (hops) displayed in the output.
5. Counted the numbered hop entries shown in the traceroute result.

Result:

The traceroute to www.youtube.com took **[06] hops** to reach the destination.

Screenshot:

```
C:\Users\milan>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4827:400::]
over a maximum of 30 hops:

  1     3 ms     3 ms     2 ms  2406:b400:f11:e9f7:bafb:b3ff:fea2:4a1b
  2    27 ms    27 ms    28 ms  2400:3300:1111::9
  3     *        *        *      Request timed out.
  4    41 ms    42 ms    41 ms  2406:b400:8::76
  5    41 ms    41 ms    42 ms  2001:4860:1:1::19bc
  6    42 ms    42 ms    41 ms  2001:4860:4827:400::

Trace complete.
```

Conclusion:

The tracert command was successfully used to identify the network path between the laptop and YouTube and determine the number of hops along the route.

Task 3 : Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Practical: Ping Test After Disconnecting a Device

Objective:

To observe the result of a ping test when the target device is disconnected from the Wi-Fi network.

Procedure:

1. Connected the laptop and phone to the same Wi-Fi network.
2. Confirmed the phone's IP address.
3. Disconnected the phone from the Wi-Fi network.
4. On the laptop, opened **Command Prompt**.
5. Tried to ping the phone using its previous IP address:

ping 192.168.0.4

6. Observed the response displayed by Command Prompt.

Error Message:

Request timed out.

Destination host unreachable.

```
C:\Users\milan>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.0.11: Destination host unreachable.
Reply from 192.168.0.11: Destination host unreachable.

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
```

Result:

After the phone was disconnected from Wi-Fi, the laptop could no longer communicate with it. The ping request failed and displayed "**Request timed out.**"

Task 4 : Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Practical: Identifying Devices on Home Wi-Fi

Objective:

To identify the IP addresses of devices communicating on the home Wi-Fi network using the arp -a command.

Command Used:

arp -a

Procedure:

1. Connected the laptop to the home Wi-Fi network.
2. Opened **Command Prompt** on Windows.
3. Entered the **arp -a** command.
4. Recorded the IP addresses and corresponding MAC addresses displayed in the ARP table.
5. Identified the devices by comparing their IP addresses/MAC addresses with the devices connected to the Wi-Fi.

Device IP Address Table

```
C:\Users\milan>arp -a
```

Interface: 192.168.0.11 --- 0x8		
Internet Address	Physical Address	Type
192.168.0.1	b8-fb-b3-a2-4a-1b	dynamic
192.168.0.4	26-da-60-52-6f-5b	dynamic
192.168.0.5	b8-fb-b3-a2-45-ad	dynamic
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Result:

The arp -a command displayed the IP and MAC addresses available in the laptop's ARP table. These addresses were used to identify the devices connected to or recently communicating through the home network.

Conclusion:

The ARP table can be used to view local network IP-to-MAC address mappings and help identify devices communicating on the Wi-Fi network.

Task 5 : Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Home Wi-Fi Network Setup and Connectivity Testing Report

Introduction

The purpose of this practical was to set up and examine a small home Wi-Fi network and verify communication between connected devices. A Windows laptop and an Android smartphone were connected to the same Wi-Fi network. Different network commands such as ipconfig, ping, tracert, and arp -a were used to check IP addressing, connectivity, network path, and devices visible on the local network.

1. Network Setup

First, the laptop and smartphone were connected to the **same Wi-Fi network**. The IP address of each device was checked from its network settings. On the Windows laptop, the ipconfig command was used to view the laptop's IPv4 address and other network information.

Command used:

Ipconfig

Screenshot:

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2406:b400:f11:e9f7:1b00:f970:c3a8:329b
    Temporary IPv6 Address. . . . . : 2406:b400:f11:e9f7:5141:1b33:3d3e:3ad0
    Link-local IPv6 Address . . . . . : fe80::7c57:b381:11:51d%8
    IPv4 Address. . . . . : 192.168.0.11
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::bafb:b3ff:fea2:4a1b%8
                               192.168.0.1
```

2. Connectivity Testing Using Ping

After identifying the IP addresses, connectivity between the laptop and smartphone was tested using the ping command. From the laptop, the phone's IP address was pinged.

ping 192.168.0.4

The phone was also used to ping the laptop's IP address. Both tests returned successful **Reply** messages with response times. This confirmed that the two devices could communicate with each other over the local Wi-Fi network.

Screenshot:

```
C:\Users\milan>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:
Reply from 192.168.0.4: bytes=32 time=77ms TTL=64
Reply from 192.168.0.4: bytes=32 time=75ms TTL=64
Reply from 192.168.0.4: bytes=32 time=3ms TTL=64
Reply from 192.168.0.4: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 77ms, Average = 39ms
```

Screenshot:

```
10:06 >_ 3.63 KB/s 84%

~ $ ping -c 5 192.168.0.11
PING 192.168.0.11 (192.168.0.11) 56(84) bytes of data:
64 bytes from 192.168.0.11: icmp_seq=1 ttl=128 time=12.6 ms
64 bytes from 192.168.0.11: icmp_seq=2 ttl=128 time=10.7 ms
64 bytes from 192.168.0.11: icmp_seq=3 ttl=128 time=11.9 ms
64 bytes from 192.168.0.11: icmp_seq=4 ttl=128 time=12.0 ms
64 bytes from 192.168.0.11: icmp_seq=5 ttl=128 time=10.9 ms

--- 192.168.0.11 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 10.733/11.673/12.688/0.739 ms
~ $
```

3. Traceroute Testing

The `tracert` command was then used on the Windows laptop to observe the network path to YouTube.

`tracert www.youtube.com`

The command displayed a list of network hops between the laptop and the destination. The traceroute took **[05] hops** to reach the destination.

Screenshot:

```
C:\Users\milan>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4827:400::]
over a maximum of 30 hops:

  1     3 ms     3 ms     2 ms  2406:b400:f11:e9f7:bafb:b3ff:fea2:4a1b
  2    27 ms    27 ms    28 ms  2400:3300:1111::9
  3     *        *        *      Request timed out.
  4    41 ms    42 ms    41 ms  2406:b400:8::76
  5    41 ms    41 ms    42 ms  2001:4860:1:1::19bc
  6    42 ms    42 ms    41 ms  2001:4860:4827:400::

Trace complete.
```

4. Testing After Disconnecting a Device

To observe a connectivity failure, the smartphone was disconnected from the Wi-Fi network. The laptop then attempted to ping the phone using its previous IP address.

ping 192.168.0.4

The ping failed and displayed:

Request timed out.

Destination host unreachable.

This showed that the laptop could no longer communicate with the phone after it was disconnected from the network.

```
C:\Users\milan>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.0.11: Destination host unreachable.
Reply from 192.168.0.11: Destination host unreachable.

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
```

5. Checking Devices Using ARP

Finally, the arp -a command was used on the laptop to view IP-to-MAC address mappings available in the local ARP table.

arp -a

The displayed IP and MAC addresses were recorded and compared with the known devices on the home network.

Screenshot:

```
C:\Users\milan>arp -a

Interface: 192.168.0.11 --- 0x8
    Internet Address      Physical Address      Type
    192.168.0.1           b8-fb-b3-a2-4a-1b    dynamic
    192.168.0.4           26-da-60-52-6f-5b    dynamic
    192.168.0.5           b8-fb-b3-a2-45-ad    dynamic
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

Results and Conclusion

The practical was completed successfully. The laptop and smartphone were able to communicate when connected to the same Wi-Fi network, as confirmed by successful ping replies from both sides. The tracert command showed the network path to YouTube and the number of hops involved. When the phone was disconnected, the ping failed with **"Request timed out"**, demonstrating that a disconnected device cannot be reached through the local network. The arp -a command also provided useful information about IP and MAC address mappings of devices visible to the laptop.